

Policy Superposition for ARC-AGI 3

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Abstract

This paper presents a preliminary evaluation of Policy Superposition, a lightweight test-time control method I co-developed with OpenAI Codex for interactive language-model agents. Instead of committing early to one interpretation of an unfamiliar environment, the agent maintains multiple candidate policies in persistent memory, revises them using real action outcomes, and decides when the evidence justifies selecting one. All candidates share one real trajectory and are updated inside ordinary action calls; the method adds no auxiliary model calls, search tree, environment fork, rollback, voting procedure, or external hypothesis manager. I evaluated the method on all 25 public ARC-AGI-3 games using DeepSeek V4 Flash and the Tufa Labs Duck harness. A matched baseline scored 6.01%, while Policy Superposition scored 11.02%, an increase of 83.4%. The treatment completed 52 levels across 24 games, compared with 42 levels across 21 games for the baseline, while using slightly fewer actions. Because this result comes from one stochastic paired evaluation with important confounds, I present it as promising preliminary evidence rather than a settled performance estimate.

1. Introduction

ARC-AGI-3 requires agents to explore unfamiliar interactive environments, infer their mechanics and objectives, maintain useful memory, and act efficiently.^[1] Early observations often support several explanations. A language-model agent may nevertheless commit to one tentative explanation and interpret later evidence through it, even when another explanation would produce better actions.

Policy Superposition is designed to delay this premature commitment. At the beginning of every level, the model maintains several meaningfully distinct candidate policies. A candidate policy is a provisional strategy for selecting actions, usually grounded in an interpretation of the environment. The model revises its candidates as real evidence arrives and may select one whenever it believes the evidence is sufficient.

Maintaining and revising multiple hypotheses is not itself a new idea. The contribution examined here is a deliberately lightweight realization for an interactive language-model agent: multiple natural-language policies coexist inside the actor's ordinary persistent memory, receive evidence from the same real trajectory, and are managed by the actor without a separate search process, hypothesis registry, world-model builder, or additional model request.

In one paired public evaluation, the mean score rose from 6.01% to 11.02%. This is approximately 1.83 times the baseline score. The title describes that observed run; repeated matched evaluations are still needed to estimate the method's expected performance.

2. Method

2.1 Candidate Policies

I use **policy** broadly and operationally to mean a provisional action-selection strategy. A candidate must imply what evidence to seek, what action to prefer, or how future actions should be chosen, even when it is grounded in a world hypothesis. Candidate policies may therefore take the form of alternative world hypotheses, action rules, investigation strategies, plans, or combinations of these. They are written in natural language inside the agent's persistent memory. After each real action, the model compares the observed effect with its candidates and may strengthen, weaken, revise, split, merge, add, or remove them. The term **superposition** is descriptive: it refers to temporarily preserving several candidates and does not denote a quantum-mechanical state.

The controller enforces only a simple lifecycle:

1. Every level begins with Policy Superposition active.
2. The model decides how many candidates are useful and how to represent them.
3. Real observations and action effects are used to update the candidates.
4. The model decides when the evidence is sufficient to select one policy.
5. Selection remains in effect for the rest of that level.
6. The next level resets Policy Superposition to active.

There is no prescribed candidate count, policy taxonomy, confidence threshold, action threshold, or mandatory selection point. The controller tracks only the level boundary and whether superposition is active or ended.

2.2 One Real Trajectory

Policy Superposition does not simulate parallel environments. The agent executes one real action at each step, and every candidate receives the same resulting evidence. There is no branch cloning, rollback, voting, or separate proposal call.

This distinguishes the method from explicit search. It adds context to ordinary action prompts, but it does not add model requests. The model reasons over several candidates while following one unavoidable real trajectory.

This branchless design is why I describe the system as lightweight. It changes the organization of the actor's memory and reasoning rather than introducing a second inference loop.

2.3 Lightweight Tufa and DeepSeek Integration

I implemented Policy Superposition as a notebook-level prompt addendum to the shareable Tufa Labs Duck harness.^[2] Tufa's Python tool, persistent world model, action loop, timeout, game lifecycle, and concurrency remained unchanged.

Both evaluated systems used DeepSeek V4 Flash through the official DeepSeek API with thinking enabled, maximum reasoning effort, a maximum output setting of 384,000 tokens, and concurrency 28.^[3] Because the model was text-only in this setup, every frame was converted deterministically into an exact letter-coded grid with a color legend.

Persistent-memory updates were required to appear in visible assistant content before tool calls. Hidden reasoning was not copied into persistent memory. Policy Superposition added 2,259,364 characters across 2,086 ordinary Tufa prompts, but it made zero auxiliary model calls.

2.4 Qualitative Transcript Example

The following abridged visible outputs come from one treatment transcript for the public game `ar25-0c556536`. They illustrate the mechanism's behavior, rather than providing additional performance evidence.

First, the model retained multiple explanations instead of immediately committing to one:

Goal model: Candidates: (a) move lock onto Y door (key to rows 45-53, cols 3-11); (b) key must cross the S wall to the Y door; (c) some SPACE/ACTION7 confirm action needed at alignment.

After entering the next level, it treated an unexpected action outcome as evidence against the rule transferred from the previous level:

World model: Level 2 LEFT press produced a surprising result: the c object moved LEFT 6 cells (18,15-29,29 -> 18,9-29,23), the B object stayed put, and ~168 S cells changed (wall appears to have shifted left 3). The Y door stayed. This contradicts level 1's simple mirror rule - need exact positions.

Later, it corrected a mistaken shape interpretation and recombined the surviving geometric relationships into a new plan:

World model: CRITICAL SHAPE CORRECTION: node20's head is at rows 51-53 (not 42-44) [...] Correct pairing: node4 (head-bottom) covers node20 directly; node7 (head-top) covers node19 directly; and with the WALL AT ROWS 27-29 (axis row 28), their mirrors c1/c2 land EXACTLY on the top doors node2/node3. This covers ALL FOUR doors simultaneously [...]

The action sequence derived from the third update returned `level_completed: True`. These excerpts show candidate preservation, evidence-driven rejection of an inherited rule, and later synthesis within one persistent trajectory. They do not establish that Policy Superposition caused the completion.

3. Relationship to Prior Work

Policy Superposition does not introduce the general idea of maintaining competing hypotheses. Its intended contribution is a lightweight, single-agent, branchless test-time protocol for interactive language-model agents: several natural-language policies share one real trajectory, are revised within ordinary action calls, and require no auxiliary inference service.

Hypothetical Minds generates natural-language hypotheses about other agents' strategies, evaluates their predictions, and iteratively refines them inside a modular multi-agent architecture.^[4] WorldLLM similarly conditions an LLM world model on natural-language hypotheses and refines them using Bayesian inference, a second LLM, and curiosity-driven reinforcement learning.^[5] These systems establish strong precedent for preserving and revising explicit natural-language hypotheses. Policy Superposition applies the general principle to action-selection policies in a much smaller scaffold: one acting model manages the candidates inside its existing memory and calls.

The Hypothesis Evolution Protocol externalizes hypotheses as persistent registry objects with evidence, belief values, lineage, and lifecycle states.^[6] It is more structured and auditable than Policy Superposition. My method leaves candidate representation, confidence, revision, and selection to the acting model, trading formal auditability for low implementation and inference overhead.

DreamTeam, introduced with Workspace Optimization, is especially relevant because it also targets ARC-AGI-3.^[7] Its multi-agent workspace stores programs, hypotheses, abandoned candidates, unresolved questions, and trace evidence, while specialized roles build and test an executable world model. Tycho is another ARC-AGI-3 system that constructs, verifies, and plans with programmatic world models; it can record competing hypotheses in notes but selects one executable hypothesis for planning.^[8] Both systems provide substantially richer modeling and verification machinery. Policy Superposition instead tests whether a useful part of hypothesis plurality can be obtained through a prompt-and-memory protocol layered onto an existing single-agent harness.

AIX describes Seed IQ as maintaining a continuous belief landscape that remains unresolved until constraints force a stable solution.^[9] This is conceptually close to delaying commitment, but the implementations are substantially different. Seed IQ is publicly described as a physics-grounded continuous inference system. Policy Superposition is a natural-language control protocol for an existing language-model agent. I did not use Seed IQ source code or an implementation of its proprietary system.

Tree of Thoughts explores explicit reasoning branches and supports evaluation, lookahead, and backtracking.^[10] Policy Superposition instead retains several policies inside one persistent interaction history. It neither constructs a tree nor executes separate branches.

Self-consistency samples independent reasoning paths and selects a final answer by agreement.^[11] Policy Superposition keeps alternatives inside one evolving agent state, where they can influence one another and change after every action. A different prior method named Superposition Prompting processes retrieved documents through parallel prompt paths for retrieval-augmented generation.^[12] I use the name **Policy Superposition** to distinguish this interactive method from that earlier RAG technique.

The underlying interaction pattern is related to ReAct, which interleaves reasoning, actions, and observations.^[13] Tufa Duck provides that practical agent loop for this experiment. Policy Superposition changes what the persistent memory preserves while the loop runs. Unlike the systems above, it does not add a dedicated hypothesis tool, separate model role, executable simulator, or extra inference call.

4. Evaluation and Results

I ran one stochastic pass over the same 25 public games for each condition:

- Baseline: [Kaggle notebook](#) and [ARC-AGI-3 scorecard](#)
- Policy Superposition: [Kaggle notebook](#) and [ARC-AGI-3 scorecard](#)

Metric	Baseline	Policy Superposition
Mean score	6.01%	11.02%
Completed levels	42	52
Games completing at least one level	21	24
Total actions	6,523	6,484
Total reported tokens	7,706,572	8,409,852

Policy Superposition scored higher on 15 games, lower on 7, and tied on 3. Two games produced especially large gains: ft09-0d8bbf25 improved by 47.02 percentage points and lp85-305b61c3 improved by 41.18 points. Together, they account for approximately 70% of the summed score difference. Excluding both leaves a mean advantage of approximately 1.62 points over the remaining 23 games.

4.1 API Incident and Sensitivity Analysis

Both runs encountered the same temporary API-balance outage. The relevant question is not whether the outage had zero effect, which cannot be known, but whether it affected the two conditions differently enough to explain the final score gap.

The logs strongly indicate matched exposure. Their first HTTP 402 retries occurred 1.8 seconds apart, their last occurred 1.0 second apart, and each retry disruption lasted approximately 10 minutes 45 seconds. After grouping retries into analyzer calls and removing duplicated console messages, every game in each condition had exactly four analyzer calls exhaust all eight attempts: 100 failed analyzer calls per run. All 25 games in both conditions subsequently recorded successful model calls after the balance was restored.

Incident check	Baseline	Policy Superposition
Analyzer calls exhausted by HTTP 402	100	100
Exhausted calls per game	4	4
Games with successful calls after recovery	25/25	25/25

Incident check	Baseline	Policy Superposition
Last printed mean before exhausted-call messages	5.26%	8.51%
Final mean	6.01%	11.02%

Policy Superposition therefore already led by 3.25 percentage points immediately before the first analyzer calls exhausted their retry budgets. After service resumed, both runs continued, and the final lead was 5.02 points. The treatment also completed 10 more levels, progressed in three more games, won the paired per-game comparison 15 to 7 with three ties, and used 39 fewer actions overall. The difference is consequently not confined to a larger action allowance after recovery.

The baseline recorded one additional non-quota timeout near the end of `ar25-0c556536`. Removing that entire game, rather than trying to guess the timeout's effect, leaves a 4.60-point treatment advantage across the other 24 games. Removing the two largest treatment gains leaves a 1.62-point advantage across 23 games. Removing those two games and `ar25-0c556536` together still leaves a 1.01-point advantage across the remaining 22 games.

These checks do not reconstruct the missing calls or prove that the outage was harmless. A single call can redirect an interactive trajectory, and the evaluation remains stochastic. They do show that attributing the full 5.02-point gap to the API incident would require a strongly asymmetric counterfactual despite identical observed 402 exposure, a treatment lead before exhausted calls, recovery in every game, and a residual advantage under conservative exclusions. I therefore treat the outage as a limitation on precision, not as an explanation that nullifies the observed comparison.

The generated mechanism report observed 77 levels. Every level began with Policy Superposition active. The model ended it on 6 levels, while 71 remained active at their final captured state. Forty-nine levels were cleared while superposition remained active, and three were cleared after it ended. No auxiliary model calls were made.

5. Interpretation

The result is consistent with the idea that preserving several plausible strategies can reduce damage from early false certainty. One real action may test several candidate explanations at once, allowing the model to gain information without separately executing every policy.

The system's practical appeal is its low structural overhead. It asks an existing action agent to organize uncertainty differently rather than adding search workers, environment copies, a formal belief database, or a learned exploration policy. This simplicity also limits auditability and control: the model may maintain weakly distinct candidates, update them inconsistently, or fail to use them.

The model rarely chose to end superposition. Most successful levels were cleared while it remained active. If the effect replicates, the important component may therefore be sustained structured plurality rather than the final selection of one policy.

Other explanations remain possible. The prompt may simply encourage less dogmatic reasoning, better-organized memory, or more frequent reconsideration. The present experiment does not isolate these effects.

6. Limitations

This report has several important limitations:

1. It contains only one paired run and does not measure run-to-run variance.
2. Both runs were disrupted by an API-balance outage.
3. Two games account for most of the aggregate improvement.
4. The treatment used approximately 9.1% more reported tokens despite using no extra calls.
5. The treatment logged 120 visible-memory contract violations, 112 of them in one game; the baseline logged 16.

6. Instrumentation can confirm the lifecycle but cannot prove that every active state contained genuinely distinct policies.
7. The intervention combines candidate maintenance, evidence-based revision, status reporting, and model-decided selection without ablations.
8. The evaluation covers only the 25 public ARC-AGI-3 games.

Repeated matched runs and component ablations are required before making a general performance claim.

7. Acknowledgments

Policy Superposition was co-developed by me and OpenAI Codex.^[14]

8. Conclusion

Policy Superposition asks an interactive agent to preserve multiple candidate policies until evidence supports commitment. In one paired evaluation over all 25 public ARC-AGI-3 games, adding the method to DeepSeek V4 Flash raised the mean score from 6.01% to 11.02%, completed 10 more levels, and progressed in three more games without adding model requests or environment branches.

The result is preliminary, but it is large enough to justify replication. A lightweight, branchless, persistent, model-managed set of candidate policies may offer a practical middle ground between early single-policy commitment and more elaborate hypothesis-management or explicit-search systems.

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